

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent
Kind Code
Date of Patent
Inventor(s)

12402495
B2
August 26, 2025
Wang; Yu et al.

Display apparatus having flexible panel and shielded multiplexer in non-display sub-region and method for manufacturing the same

Abstract

A display apparatus and a method for manufacturing the display apparatus are provided. The display apparatus includes: a flexible panel which includes a display region and a non-display region, where the non-display region includes a bending sub-region and a back sub-region connected to the bending sub-region and arranged opposite to the display region, a multiplexer is arranged in the back sub-region; and a shielding electrode layer located on a side of the multiplexer away from the back sub-region, an orthographic projection of the multiplexer onto a plane where the shielding electrode layer is located being within the shielding electrode layer.

Inventors:	Wang; Yu (Beijing, CN), Han; Linhong (Beijing, CN), Liu; Tingliang (Beijing, CN), Zhang; Yi (Beijing, CN), Yang; Huijuan (Beijing, CN), Shang; Tinghua (Beijing, CN), Zhou; Yang (Beijing, CN), Yu; Pengfei (Beijing, CN), Zhang; Shun (Beijing, CN), Zhang; Hao (Beijing, CN), Jiang; Xiaofeng (Beijing, CN), Li; Huijun (Beijing, CN)
Applicant:	Chengdu BOE Optoelectronics Technology Co., Ltd. (Sichuan, CN); BOE Technology Group Co., Ltd. (Beijing, CN)
Family ID:	1000008782763
Assignee:	Chengdu BOE Optoelectronics Technology Co., Ltd. (Sichuan, CN); BOE Technology Group Co., Ltd. (Beijing, CN)
Appl. No.:	17/438395
Filed (or PCT Filed):	February 09, 2021
PCT No.:	PCT/CN2021/076178
PCT Pub. No.:	WO2021/164639
PCT Pub. Date:	August 26, 2021

Prior Publication Data

Document Identifier

US 20220384557 A1

Publication Date

Dec. 01, 2022

Foreign Application Priority Data

CN

202010102912.5

Feb. 19, 2020

Publication Classification

Int. Cl.: **H10K59/126** (20230101); **H10K59/131** (20230101); H10K59/12 (20230101); H10K71/00 (20230101); H10K102/00 (20230101)

U.S. Cl.:

CPC **H10K59/126** (20230201); **H10K59/131** (20230201); H10K59/1201 (20230201); H10K71/00 (20230201); H10K2102/311 (20230201)

Field of Classification Search

CPC: H10K (59/131); H10K (71/00); H10K (59/1201); H10K (2102/311); H10K (59/126); H04M (1/0268); H04M (1/0269); H01Q (1/22); H01Q (1/526); H01Q (1/2283); H01L (23/552); H01L (23/66); H01L (2223/6677)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2013/0016065	12/2012	Reynolds	345/174	H05K 1/0306
2015/0294987	12/2014	Xia	257/659	H01L 23/552
2015/0382446	12/2014	Kwon	174/251	G06F 1/1643
2018/0183910	12/2017	Wu	N/A	H04B 5/26
2019/0033493	12/2018	Kwon et al.	N/A	N/A
2020/0403597	12/2019	Takeuchi	N/A	H03H 7/463
2021/0181888	12/2020	Yan et al.	N/A	N/A
2021/0192987	12/2020	Du et al.	N/A	N/A
2021/0225982	12/2020	Yi	N/A	N/A
2022/0164002	12/2021	An	N/A	G06F 1/1652

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
103926732	12/2013	CN	N/A
109326628	12/2018	CN	N/A
110187797	12/2018	CN	N/A
110570772	12/2018	CN	N/A
110992834	12/2019	CN	N/A
111312765	12/2019	CN	N/A
2019215726	12/2018	JP	N/A

OTHER PUBLICATIONS

Translation of JP-2019215726-A (Year: 2019). cited by examiner

CN 2020101029125 first office action. cited by applicant

PCT/CN2021/076178 international search report and written opinion. cited by applicant

Primary Examiner: Monbleau; Davienne N

Assistant Examiner: Cornely; John Patrick

Attorney, Agent or Firm: IPro, PLLC

Background/Summary**CROSS-REFERENCE TO RELATED APPLICATIONS**

(1) The present application claims a priority to Chinese Patent Application No. 202010102912.5 filed in China on Feb. 19, 2020, which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

(2) The present disclosure relates to the field of display technology, in particular to a display apparatus and a method for manufacturing the display apparatus.

BACKGROUND

(3) An active-matrix organic light-emitting diode (AMOLED) display technology has developed rapidly, has a broad application prospect, and has been widely applied in mobile products.

SUMMARY

(4) The present disclosure provides a display apparatus, including: a flexible panel, where the flexible panel includes a display region and a non-display region, the non-display region includes a bending sub-region and a back sub-region connected to the bending sub-region and arranged opposite to the display region, and a multiplexer is arranged in the back sub-region; and a shielding electrode layer located on a side of the multiplexer away from the back sub-region, an orthographic projection of the multiplexer onto a plane where the shielding electrode layer is located being within the shielding electrode layer.

(5) Optionally, the display apparatus further includes a power supply line located on the side of the multiplexer away from the back sub-region and connected to the shielding electrode layer.

(6) Optionally, a plurality of display signal lines is further arranged in the back sub-region, and at least one display signal line is further used as the power supply line.

(7) Optionally, a voltage inputted into the display signal line is a direct-current voltage.

(8) Optionally, the display signal line includes a VDD signal line further used as the power supply line.

(9) Optionally, a first insulation layer is provided between the shielding electrode layer and the VDD signal line, and the shielding electrode layer is connected to the VDD signal line through a first via-hole penetrating the first insulation layer.

(10) Optionally, the display signal line includes a VSS signal line further used as the power supply line.

(11) Optionally, a second insulation layer is provided between the shielding electrode layer and the VSS signal line, and the shielding electrode layer is connected to the VSS signal line through a second via-hole penetrating the second insulation layer.

(12) Optionally, the display signal line includes a VGH signal line further used as the power supply

- line.
- (13) Optionally, a third insulation layer is provided between the shielding electrode layer and the VGH signal line, and the shielding electrode layer is connected to the VGH signal line through a third via-hole penetrating the third insulation layer.
- (14) Optionally, the display signal line includes a VGL signal line further used as the power supply line.
- (15) Optionally, a fourth insulation layer is provided between the shielding electrode layer and the VGL signal line, and the shielding electrode layer is connected to the VGL signal line through a fourth via-hole penetrating the fourth insulation layer.
- (16) Optionally, the display apparatus further includes: a chip on film arranged in the back sub-region, where a ground line is provided on the chip on film, and the ground line is further used as the power supply line.
- (17) Optionally, the shielding electrode layer is connected to the ground line through a connection line.
- (18) Optionally, the display apparatus further includes a bearing structure arranged opposite to the flexible panel and an antenna unit located on the bearing structure, and both the bearing structure and the antenna unit are arranged on one side of the shielding electrode layer away from the back sub-region.
- (19) Optionally, an orthographic projection of the antenna unit onto the plane where the shielding electrode layer is located is within the shielding electrode layer.
- (20) Optionally, the display region of the flexible panel includes a plurality of organic light-emitting diode (OLED) display units.
- (21) The present disclosure further provides a method for manufacturing the above-mentioned display apparatus, including: forming the flexible panel, and forming the display region and the non-display region on the flexible panel, where the non-display region includes the bending sub-region and the back sub-region connected to the bending sub-region and arranged opposite to the display region; mounting the multiplexer in the back sub-region; and forming the shielding electrode layer at the side of the multiplexer away from the back sub-region, where the orthographic projection of the multiplexer onto the plane where the shielding electrode layer is located is within the shielding electrode layer.
- (22) Optionally, when the ground line provided on the chip on film is further used as a power supply line for applying a stable voltage to the shielding electrode layer, the method further includes: after the shielding electrode layer is formed, arranging a connection line between the shielding electrode layer and the ground line of the chip on film, to connect the shielding electrode layer to the power supply line.
- (23) Optionally, the flexible panel includes a base substrate and an insulation layer on the base substrate, a plurality of display signal lines is further arranged in the back sub-region, and when at least one display signal line is further used as the power supply line, the forming the shielding electrode layer on the side of the multiplexer away from the back sub-region includes: forming a via-hole penetrating the insulation layer at a position, corresponding to the display signal line, on the insulation layer, away from the base substrate, in the back sub-region; and depositing a conductive material on the insulation layer provided with the via-hole to form the shielding electrode layer, where the shielding electrode layer is connected to the display signal line through the via-hole.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a sectional view of a display apparatus according to an embodiment of the present

disclosure;

(2) FIG. 2 is a planer view of the display apparatus according to an embodiment of the present disclosure;

(3) FIG. 3 is another planer view of the display apparatus according to an embodiment of the present disclosure;

(4) FIG. 4 is yet another planer view of the display apparatus according to an embodiment of the present disclosure;

(5) FIG. 5 is a sectional view of the display apparatus at a non-display region according to an embodiment of the present disclosure;

(6) FIG. 6 is a flow chart illustrating a method for the manufacturing display apparatus according to an embodiment of the present disclosure; and

(7) FIG. 7 is a schematic diagram of an operation principle of an MUX according to an embodiment of the present disclosure.

DETAILED DESCRIPTION

(8) In order to make the technical problems to be solved, technical solutions and advantages of the present disclosure more apparent, a detailed description will be given below with reference to the accompanying drawings and specific embodiments.

(9) In an AMOLED display apparatus, in order to save a space occupied by a driving part, improve a screen-to-body ratio and achieve a full screen display effect, it is necessary to bend a part of a driving region of a display panel to a back side of the display panel during an assembly, and a driving member is arranged in the driving region bent to the back side of the display panel. In the AMOLED display apparatus of the related art, the driving member includes a multiplexer (MUX) for applying a voltage control signal to the display panel.

(10) A voltage signal inputted into the MUX alternates between a high voltage and a low voltage, and the alternating voltage signal may form electromagnetic radiation. A circuit board is arranged inside the display apparatus and opposite to the display panel, and an overall antenna is arranged in the circuit board. The electromagnetic radiation generated by the MUX may generate electromagnetic interference to the overall antenna, so as to affect a usage effect of the overall antenna.

(11) In order to solve the problem that, in the display apparatus of the related art, the electromagnetic interference is caused when a part of the driving region of the display panel is bent to the back side of the display panel, and the alternating voltage signal of the MUX arranged in the driving region generates electromagnetic radiation. The embodiments of the present disclosure provide a display apparatus, a non-display region on a flexible panel of the display apparatus includes a back sub-region bent to the back side of the display region, a multiplexer (MUX) is arranged in the back sub-region, and a shielding electrode layer is arranged in the back sub-region, and covers the MUX, so as to shield the electromagnetic interference generated by the MUX.

(12) The display apparatus according to an embodiment of the present disclosure, as shown in FIGS. 1 and 2, includes: a flexible panel **100**, where the flexible panel **100** includes a display region **110** and a non-display region **120**, the non-display region **120** includes a bending sub-region **121** and a back sub-region **122** connected to the bending sub-region **121** and arranged opposite to the display region **110**, and a multiplexer MUX **200** arranged in the back sub-region **122**; and a shielding electrode layer **300** located on a side of the MUX **200** away from the back sub-region **122**, an orthographic projection of the MUX **200** onto a plane where the shielding electrode layer **300** is located being within the shielding electrode layer **300**.

(13) In some embodiments, the display apparatus further includes a power supply line located on the side of the MUX **200** away from the back sub-region **122** and connected to the shielding electrode layer **300**.

(14) In the embodiment of the present disclosure, the power supply line is connected to the shielding electrode layer **300** and used to apply a voltage to the shielding electrode layer **300**, so

that the shielding electrode layer **300** has a stable potential. Since the shielding electrode layer **300** is arranged on the side of the MUX **200** away from the back sub-region **122**, and the orthographic projection of the MUX **200** onto the plane where the shielding electrode layer **300** is located is within the shielding electrode layer **300**, the shielding electrode layer **300** covers the MUX **200** on the side of the MUX **200** away from the back sub-region **122**, and the shielding electrode layer **300** having a stable potential absorbs and reflects the electromagnetic interference generated by the MUX **200**. Thus, the electromagnetic interference generated by the MUX **200** is confined to a side of the shielding electrode layer **300** toward the MUX **200**, and prevented from diffusing to a side of the shielding electrode layer **300** away from the MUX **200**.

(15) In one embodiment of the present disclosure, as shown in FIG. **1**, the display apparatus further includes a bearing structure **400** arranged opposite to the flexible panel **100** and an antenna unit **500** on the bearing structure **400**; and both the bearing structure **400** and the antenna unit **500** are arranged on one side of the shielding electrode layer **300** away from the back sub-region **122**.

(16) In the embodiment of the present disclosure, since the shielding electrode layer **300** is located between the MUX **200** and the antenna unit **500**, the electromagnetic interference generated by the MUX **200** may be prevented from diffusing to the antenna unit **500** by using the shielding electrode layer **300**. Therefore, in the display apparatus according to the embodiment of the present disclosure, it is able to solve the problem that the alternating voltage signal of the MUX generates electromagnetic radiation and causes electromagnetic interference to the antenna.

(17) In the embodiment of the present disclosure, optionally, an orthographic projection of the antenna unit **500** onto the plane where the shielding electrode layer **300** is located is within the shielding electrode layer **300**.

(18) Based on this arrangement, the electromagnetic interference caused to the antenna by electromagnetic radiation generated from the alternating voltage signal of the MUX **200** can be effectively avoided by the shielding electrode layer **300**.

(19) Optionally, the bearing structure **400** for bearing the antenna unit **500** may be a flexible circuit board, or may be a support frame or the like specifically for fixing the antenna unit **500**.

(20) In the display apparatus according to an embodiment of the present disclosure, a display function device is formed on the flexible panel **100** for realizing a display function, and a region where the display function device is located on the flexible panel **100** is correspondingly formed as the display region **110**. On the flexible panel **100**, a region where the display functional device is not provided is the non-display region **120**, and a driving assembly and a chip on film **600** are arranged in the non-display region **120** located at one end of the display region **110**. The chip on film **600** is used to connect the driving assembly to the flexible circuit board for applying a driving control signal to the driving component.

(21) In addition, there is a spacing region between the driving assembly and the display region **110**, and the bending sub-region **121** is located in the spacing region.

(22) As shown in FIG. **1**, the non-display region **120** is bent in a direction opposite to the display region **110** through the bending sub-region **121**, so that the driving assembly and the chip on film **600** are bent to the back side opposite to the display region **110**. That is, a portion, opposite to the display region **110**, of the non-display region **120** forms the back sub-region **122**, and the driving assembly and the chip on film **600** are located in the back sub-region **122**.

(23) In the embodiment of the present disclosure, optionally, the display functional device in the display region **110** of the flexible panel **100** includes a plurality of organic light-emitting diode (OLED) display units.

(24) Further, optionally, the driving assembly in the back sub-region **122** includes the MUX **200**.

(25) Optionally, in the embodiment of the present disclosure, as shown in FIG. **7**, the MUX **200** may be used for the data signal distribution in the display panel. A data signal outputted by a display driving chip of the display apparatus is distributed through the MUX **200** and then inputted into the display region of the display panel, so as to reduce the number of lines for inputting the

data signal and reduce a frame space and a rounded-corner space in the non-display region.

(26) Optionally, the MUX **200** includes a plurality of MUX units, each MUX unit includes 2 or more thin film transistors TFT. In a case that each MUX unit includes two TFTs, source electrodes of the two TFTs are connected to a same data signal line inputted by the display driving chip, and drain electrodes are connected to the display region through two columns of different data signal lines. Each TFT is turned on or off through an inputted gate signal, so as to achieve the signal distribution.

(27) Based on the MUX **200** having the above structure, in each MUX unit, the gate signal is at a low level when the TFT is in an on state and at a high level when the TFT is in an off state, and within one frame, the gate signal is switched between the high level and the low level according to a certain frequency. A change in potential may be referred to as a change in current and charge, and a time-varying current may generate electromagnetic radiation, so as to cause the problem of the electromagnetic interference generated by the MUX **200**.

(28) In addition, in the display apparatus according to the embodiment of the present disclosure, in one embodiment, a ground line is provided on the chip on film **600**, and the ground line is further used as the power supply line for applying a stable voltage to the shielding electrode layer **300**.

(29) As shown in FIG. 2, the shielding electrode layer **300** is connected to the ground line through a connection line. Optionally, the shielding electrode layer **300** is connected to a ground terminal **601**, **602** of the chip on film **600** through the connection line.

(30) In the embodiment, when the shielding electrode layer **300** is connected to the ground line on the chip on film **600**, so that the electromagnetic interference generated by the MUX **200** is prevented from diffusing to the side of the shielding electrode layer **300** away from the MUX **200**, and is grounded.

(31) In the display apparatus according to an embodiment of the present disclosure, as shown in FIGS. 3 and 4, on the flexible panel, display signal line for inputting a display control signal to the display function device of the display region **110** is further arranged in the non-display region **120**. The display signal line includes a VDD signal line **701** and a VSS signal line **702**, the VDD signal line **701** includes a first portion arranged along an edge, close to the chip on film **600**, of the display region **110**, and a second portion between the chip on film **600** and the display region **110**, and used to connect the first portion to the chip on film **600**. The VSS signal line **702** includes a third portion surrounding edges of the display region **110**, and a fourth portion between the chip on film **600** and the display region **110** and used to connect the third portion to the chip on film **600**.

(32) In some embodiments, a voltage inputted into the display signal line is a direct-current voltage.

(33) In the display apparatus of the present disclosure, the VDD signal line **701** may be further used as the power supply line for applying a stable voltage to the shielding electrode layer **300**, as shown in FIG. 3.

(34) In another embodiment, the VSS signal line **702** may be further used as the power supply line for applying a stable voltage to the shielding electrode layer **300**, as shown in FIG. 4.

(35) In the embodiment, the VDD signal line **701** or the VSS signal line **702** is further used as the power supply line for applying the stable voltage to the shielding electrode layer **300**, so that the shielding electrode layer **300** connected to the VDD signal line **701** or the VSS signal line **702** has a stable potential and the electromagnetic interference generated by the MUX **200** may be prevented from diffusing to the side of the shielding electrode layer **300** away from the MUX **200**.

(36) Optionally, when the VDD signal line **701** or the VSS signal line **702** is further used as the power supply line for applying the stable voltage to the shielding electrode layer **300**, an insulation layer (i.e., a first insulation layer or a second insulation layer) is provided between the VDD signal line **701** or the VSS signal line **702** and the shielding electrode layer **300**. The shield electrode layer **300** is connected to the VDD signal line **701** or the VSS signal line **702** through a via-hole penetrating the insulation layer.

(37) As shown in FIG. 3, when the VDD signal line **701** is further used as the power supply line for applying the stable voltage to the shielding electrode layer **300**, the shielding electrode layer **300** is connected to the second portion of the VDD signal line **701** between the chip on film **600** and the display region **110** through a first via-hole **801** penetrating the first insulation layer.

(38) As shown in FIG. 4, when the VSS signal line **702** is further used as the power supply line for applying the stable voltage to the shielding electrode layer **300**, the shielding electrode layer **300** is connected to the fourth portion of the VSS signal line **702** between the chip on film **600** and the display region **110** through a second via-hole **802** penetrating the second insulation layer.

(39) In some embodiments, when the VDD signal line **701** and the VSS signal line **702** are arranged on a same layer, the first insulation layer and the second insulation layer are a same insulation layer, and the first via-hole **801** and the second via-hole **802** are via-holes at different positions of the insulation layer.

(40) In some embodiments, the display signal line may further include a VGH signal line and a VGL signal line, which may be further used as the power supply line.

(41) In some embodiments, when the VGH signal line is further used as the power supply line, a third insulation layer is provided between the shielding electrode layer and the VGH signal line, and the shielding electrode layer is connected to the VGH signal line through a third via-hole penetrating the third insulation layer.

(42) In some embodiments, when the VGL signal line is further used as the power supply line, a fourth insulation layer is provided between the shielding electrode layer and the VGL signal line, and the shielding electrode layer is connected to the VGL signal line through a fourth via-hole penetrating the fourth insulation layer.

(43) In some embodiments, when the VGH signal line and the VGL signal line are arranged on a same layer, the third insulation layer and the fourth insulation layer are a same insulation layer, and the third via-hole and the fourth via-hole are via-holes at different positions of the insulation layer.

(44) In some embodiments, the first insulation layer, the second insulation layer, the third insulation layer, and/or the fourth insulation layer may be an organic layer or an inorganic layer.

(45) As shown in FIG. 5, the non-display region **120** of the flexible panel **100** includes a base substrate **1201**, and a first inorganic layer **1202**, a second inorganic layer **1203**, a signal line layer **1204**, and an organic layer **1205** which are sequentially laminated on the base substrate **1201**, where the organic layer **1205** has a planarization function. The shielding electrode layer **300** is arranged on the organic layer **1205** formed as an insulation layer provided between the signal line layer **1204** and the shielding electrode layer **300**. A via-hole (as shown in FIG. 5, a part of the organic layer **1205** is removed to form the via-hole) is provided at a position of the organic layer **1205** corresponding to the signal line layer **1204**, and the shielding electrode layer **300** is connected to the signal line layer **1204** through the via-hole penetrating the organic layer **1205**.

(46) In some embodiments, on the flexible panel **100**, the VDD signal line **701** and the VSS signal line **702** are arranged on the same layer and separated from each other, and the signal line layer **1204** is the VDD signal line **701** or the VSS signal line **702**.

(47) In some embodiments, the VGH signal line and the VGL signal line are arranged on the same layer and separated from each other, and the signal line layer **1204** is the VGH signal line or the VGL signal line.

(48) In the display apparatus according to the above-mentioned embodiments of the present disclosure, the shielding electrode layer is arranged in the back sub-region, and covers the MUX, and the shielding electrode layer is connected to the VDD signal line, the VSS signal line, the VGH signal line, the VGL signal line or the ground line of the chip on film, so as to provide the stable voltage for the shielding electrode layer and shield the electromagnetic interference generated by the MUX.

(49) The embodiments of the present disclosure further provide a method for manufacturing the above-mentioned display apparatus, as shown in FIG. 6, in conjunction with FIGS. 1 to 5,

including the following steps.

(50) Step **S610**, forming the flexible panel **100**, and forming the display region **110** and the non-display region **120** on the flexible panel **100**, where the non-display region includes the bending sub-region and the back sub-region connected to the bending sub-region and arranged opposite to the display region.

(51) Step **S620**, mounting the MUX **200** in the back sub-region **122** of the non-display region **120**.

(52) Step **S630**, forming the shielding electrode layer **300** at the side of the MUX **200** away from the back sub-region **122**, where the orthographic projection of the MUX **200** onto the plane where the shielding electrode layer **300** is located is within the shielding electrode layer **300**.

(53) In some embodiments, the non-display region **120** includes the power supply line connected to the shielding electrode layer **300**.

(54) In one implementation, when a ground line provided on the flip chip thin film **600** is further used as the power supply line for applying a stable voltage to the shielding electrode layer **300**, after the shielding electrode layer **300** is formed in step **S630**, a connection line is arranged between the shielding electrode layer **300** and the ground line of the chip on film **600**, so as to realize a connection between the shielding electrode layer **300** and the power supply line.

(55) In some embodiments, the flexible panel includes a base substrate and an insulation layer located on the base substrate, and a plurality of display signal lines is further arranged in the back sub-region, when at least one display signal line is further used as the power supply line, the forming the shielding electrode layer on the side of the MUX away from the back sub-region includes: forming a via-hole penetrating the insulation layer at a position, corresponding to the display signal line, on the insulation layer, away from the base substrate, in the back sub-region; and depositing a conductive material on the insulation layer provided with the via-hole to form the shielding electrode layer, where the shielding electrode layer is connected to the display signal line through the via-hole.

(56) In some embodiments, the insulation layer may be an organic layer.

(57) In some embodiments, the display signal lines may be a VDD signal line, a VSS signal line, a VGH signal line or a VGL signal line.

(58) In some embodiments, when the VDD signal line **701** in the non-display region **120** is further used as the power supply line for applying the stable voltage to the shielding electrode layer **300**, in conjunction with FIG. 5, the forming the shielding electrode layer **300** in the back sub-region **122** in step **S630** includes: forming a first via-hole **801** penetrating the organic layer **1205** at a position, corresponding to the VDD signal line **701**, of the organic layer **1205**, away from the base substrate **1201**, in the back sub-region **122**; and depositing a conductive material on the organic layer **1205** provided with the first via-hole **801** to form the shielding electrode layer **300**, where the shielding electrode layer **300** is connected to the VDD signal line **701** through the first via-hole **801**.

(59) In some embodiments, when the VSS signal line **702** in the non-display region **120** is further used as the power supply line for applying the stable voltage to the shielding electrode layer **300**, in connection with FIG. 5, the forming the shielding electrode layer **300** in the back sub-region **122** in step **S630** includes: forming a second via-hole **802** penetrating the organic layer **1205** at a position, corresponding to the VSS signal line **702**, of the organic layer **1205**, away from the base substrate **1201**, in the back sub-region **122**; and depositing a conductive material on the organic layer **1205** provided with the second via-hole **802** to form the shielding electrode layer **300**, where the shielding electrode layer **300** is connected to the VSS signal line **702** through the second via-hole **802**.

(60) It should be appreciated that, an implementation of the power supply line for applying the stable voltage to the shielding electrode layer **300** in the display apparatus and the method for manufacturing the display apparatus according to the present disclosure is merely illustrative and not limited thereto, and other implementations shall be adopted by those skilled in the art based on the principles of the embodiments of the present disclosure and are not particularly defined herein.

(61) The above embodiments are optional embodiments of the present disclosure, it should be appreciated that those skilled in the art may make various improvements and modifications without departing from the principle of the present disclosure, and these improvement and modifications shall fall within the protection scope of the present disclosure.

Claims

1. A display apparatus, comprising: a flexible panel, wherein the flexible panel comprises a display region and a non-display region, the non-display region comprises a bending sub-region and a back sub-region connected to the bending sub-region and arranged opposite to the display region, and a multiplexer is arranged in the back sub-region; and a shielding electrode layer located on a side of the multiplexer away from the back sub-region, an orthographic projection of the multiplexer onto a plane where the shielding electrode layer is located being within the shielding electrode layer; wherein the display apparatus further comprises: a power supply line located on the side of the multiplexer away from the back sub-region and connected to the shielding electrode layer; wherein a plurality of display signal lines is further arranged in the back sub-region, and at least one of the plurality of display signal lines is further used as the power supply line; wherein the at least one of the plurality of display signal lines comprises a VDD signal line and a VSS signal line, and the VDD signal line or the VSS signal line is further used as the power supply line; wherein the display apparatus further comprises a chip on film arranged in the back sub-region; wherein the VDD signal line comprises a first portion arranged along an edge, adjacent to the chip on film, of the display region, and a second portion between the chip on film and the display region, and used to connect the first portion to the chip on film; wherein the VSS signal line comprises a third portion surrounding edges of the display region, and a fourth portion between the chip on film and the display region and used to connect the third portion to the chip on film; wherein a first insulation layer is provided between the shielding electrode layer and the VDD signal line, a second insulation layer is provided between the shielding electrode layer and the VSS signal line, the VDD signal line and the VSS signal line are arranged on a same layer, and the first insulation layer and the second insulation layer are a same insulation layer; wherein in a case that the VDD signal line is further used as the power supply line, an orthographic projection of the second portion of the VDD signal line onto the back sub-region at least overlaps an orthographic projection of the shielding electrode layer onto the back sub-region, and the shielding electrode layer is connected to the second portion of the VDD signal line through a first via-hole penetrating the first insulation layer; wherein in a case that the VSS signal line is further used as the power supply line, an orthographic projection of the fourth portion of the VSS signal line onto the back sub-region at least overlaps the orthographic projection of the shielding electrode layer onto the back sub-region, and the shielding electrode layer is connected to the fourth portion of the VSS signal line through a second via-hole penetrating the second insulation layer.

2. The display apparatus according to claim 1, further comprising a bearing structure arranged opposite to the flexible panel and an antenna unit located on the bearing structure, wherein both the bearing structure and the antenna unit are arranged on one side of the shielding electrode layer away from the back sub-region.

3. The display apparatus according to claim 2, wherein an orthographic projection of the antenna unit onto the plane where the shielding electrode layer is located is within the shielding electrode layer.

4. The display apparatus according to claim 1, wherein the display region of the flexible panel comprises a plurality of organic light-emitting diode (OLED) display units.

5. A method for manufacturing the display apparatus according to claim 1, comprising: forming the flexible panel, and forming the display region and the non-display region on the flexible panel, wherein the non-display region comprises the bending sub-region and the back sub-region

connected to the bending sub-region and arranged opposite to the display region; mounting the multiplexer in the back sub-region; and forming the shielding electrode layer at the side of the multiplexer away from the back sub-region, wherein the orthographic projection of the multiplexer onto the plane where the shielding electrode layer is located is within the shielding electrode layer.

6. The method according to claim 5, wherein the flexible panel comprises a base substrate and an insulation layer on the base substrate, the plurality of display signal lines is further arranged in the back sub-region, and when at least one of the plurality of display signal lines is further used as the power supply line, the forming the shielding electrode layer on the side of the multiplexer away from the back sub-region comprises: forming a via-hole penetrating the insulation layer at a position, corresponding to the at least one of the plurality of display signal lines, on the insulation layer, away from the base substrate, in the back sub-region; and depositing a conductive material on the insulation layer provided with the via-hole to form the shielding electrode layer, wherein the shielding electrode layer is connected to the at least one of the plurality of display signal lines through the via-hole.
